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Solution Architect

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Experience Summary

I hold a **B.E.** degree in **Computer Science** and Engineering and carry over **17 years** of extensive experience in software development, testing, and consulting. Throughout my career, I've cultivated expertise across diverse sectors including telecom, travel, banking and LOS. I have worked as Technical Architect on multiple projects.

My professional journey has involved contributing to a multitude of projects spanning various products and web-based applications. My proficiency spans across a spectrum of Java/J2EE technologies.

- I have experience in following **Backend** technologies, utilizing **Java** versions 11, 17 & 21.
 - **Microservices** architecture, employing **Spring Boot** and **Spring Cloud** with a focus on Domain Driven Design (**DDD**) principles.
 - Implementation of **Reactive** microservices using Spring **Webflux** and **Reactor**.
 - **CQRS** and **ES** based microservices implementation using **AXON** Framework.
 - Hands on **Apache Kafka** ([CCDAK](#)), **Kafka Connect** and **Kafka Streams**.
 - Extensive experience in **Spring framework**, especially with modules **Core**, **AOP**, **Rest**, **Webflux**, **Data**, **JDBC**, **Batch**, **Security**, **Authorization server**, **API Gateway**, **Spring AI**.
 - Netflix **GraphQL** & Apollo GraphQL Client implementation.
 - Application security using **Spring Security**, **JWT**, **OpenID/OAuth2**, **LDAP**, **ACL**, **Keycloak**, **Azure AD** and **OWASP**.
 - Familiarity with **AWS** services such as **Step Functions**, **S3** and **DynamoDB**.
 - API management using **Swagger/OpenAPI 3.0**, **API Curio** and API Testing using **Postman**.
 - Extensive **database** knowledge covering **MongoDB**, **MySQL**, **DB2**, **Oracle**, **PostgreSQL** and implementation of **DAO** and **Repository** patterns using **SQL**, **JDBC**, **Spring Data**, **Hibernate**.
 - **Multithreading**, **Data Structures**, **Collections** framework, **Core Design patterns**, **Integration patterns (EIP)** using **Apache Camel** and **messaging** using **RebbitMQ**, **Kafka** & **ActiveMQ**.
 - **Jobrunr** & **Quartz job scheduling**, **MVEL expression engine**, **Easy Rules rule engine**.
 - **Build** using **Maven** & **Gradle** and **deployment** with **Docker**, **Kubernetes** and **Jenkins**.
 - Basic knowledge in **Python**.
- I have experience in following **Data engineering** technologies.
 - Large data processing using **Spring Batch** Jobs implementation as **Spring Cloud Task** and execution on **Spring Cloud Data flow**.
 - **Feature engineering** using **Apache Spark** and **Data Lakehouse** implementation using **Apache Iceberg** with **Hadoop**, **Hive** and **Nessie** catalogs.
 - Hands on **Feast feature store** implementation with various offline stores and its Java client implementation to manage Feast metadata, offline, online data ingestion and materialization.
- I have implemented a generic **MCP** (Model Context Protocol) server to turn any third-party REST APIs to AI-callable Tools using **Spring AI**.
- I am Expert contributor to all phases of Software Development Life Cycle (**SDLC**).
- I have expertise in writing **Unit tests** using **JUnit** and **Mockito** and **Integration testing** using **Wiremock**.
- I am passionate about staying abreast of the latest technologies, methodologies, and **best practices** and I actively engage in continuous learning and knowledge sharing through **blogging**.

Published Works

- A generic [MCP server](#) (Model Context Protocol) to turn any third-party REST APIs to AI-callable Tools.
- [Framework for Building Spark Jobs](#) with Spring boot and Running on Local and Kubernetes
- [Spring boot starter for spark](#) with dependency descriptors that you can include in your Spring boot application to have all required Spark dependencies and *SparkSession* and *Iceberg Catalog* beans auto-configured with spark configurations support in application.yml.
- [Apache spark Iceberg integration](#) using Hadoop, Hive and Nessie catalogs.
- [Spring boot starter for Feast](#) to auto-configure Feast registry, service server client beans and a **fluent client** to manage metadata, offline and online data ingestion and features materialization.

- [Spring Boot Problem Handler](#): Fully customizable exception handling library for both Spring Web (Servlet) and Spring Webflux (Reactive) Rest services. Demo for [Spring Web](#) and [Spring Webflux](#)
- [Spring Batch commons components](#): Customizable Spring batch components to create Simple and Partitioned Spring batch job easily. Demo for Spring Batch Job implemented as [Spring cloud task](#) and [Spring Rest Service](#).
- [Spring Data MongoDB Auditing](#): Hibernate Envers style Audit logging for Spring data MongoDB and Text search example.

Certifications

- Confluent Certified Developer for Apache Kafka ([CCDAK](#))

Below is a list of key projects that I have worked upon.

MLOS	Solution Architect	Apr 2023 – till date	Product
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Description: This product provides a platform for defining, executing, and deploying feature engineering and machine learning pipelines. It enables users to define feature pipelines through an intuitive UI to define

- Data sources for ingesting structured and unstructured data like files from filesystem or cloud platforms like AWS S3, SQL/NoSQL databases, Kafka etc. or any other kind.
- The transformation steps to apply on various datasets such as aggregations, join, pivot etc. for feature engineering.
- Data Lakehouse implementation using Hadoop, Hive and Apache Iceberg for cataloguing, metadata and data storage, **Apache Atlas** for **Data governance** and **Apache Ranger** for data security.

Additionally, users can define ML pipelines by specifying

- Entities, Attributes, Labels and Features to fetch.
- Preprocessing steps such as outliers, imputations etc.
- Automated experimentation, training and AutoML to identify and deploy the best-performing machine learning models for batch scoring or real-time inference.

It uses Apache Spark to process batch and streaming data, applying transformations for feature engineering and output into a scalable data lakehouse, from where offline/online features are fetched using Feast.

Responsibilities:

- Conducted extensive R&D to evaluate and adopt technologies like Spark, [Feast](#) and **Apache Iceberg**, **Nessie**, Trino, **Dremio** etc.
- Architected and developed a [unified framework](#) to build Spark jobs using Spring Boot, running locally in IDE and deployment on Kubernetes to simplify the development and deployment of Spark jobs.
- Built a Java Feast client to manage feast metadata such as entities, data source, feature view etc. programmatically from other services. Implemented Feature fetching using Python, and serving the Features to ML Model for training, batch scoring and inferencing.
- Built a Java client for Apache Iceberg Catalog management
- Ensured scalability, fault tolerance, and seamless integration with different technology stack.

Khata LMS	Technical Architect	Apr 2023 – Nov 2023	Piramal Finance
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Description: Piramal Finance provides housing finance and other financial solutions ranging wide capital stack. Khata LMS is a product to manage the whole lifecycle of a Loan from its inception, repayment schedules, apportionments, transaction management etc., reporting and auditing. It exposes the APIs to its partners to manage their Loan products.

Responsibilities:

- Architecture design and implementation of the system, ensuring scalability, reliability, and performance.
- Developed and optimized [Spring batch jobs](#) for critical financial processes such as Accrual, Days Past Due (DPD), and Late Payment Interest (LPI).
- Ensured high performance of jobs using partitioning, processing 2 million records in less than 5 minutes, and deployed these batch jobs on **Spring Cloud Dataflow**.

- Implemented **Workflow Orchestration** using **AWS Step functions** and **Spring boot microservices** for master data, charges, repayment schedules and apportionments. Inter-microservices communication using **Kafka**.

ABG DirectConnect	Technical Architect	Feb 2022 – Mar 2023	Avis Budget Group
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Description: Avis Budget Group, Inc. is a leading global provider of mobility solutions for rental vehicles. API Modernization caters to exposing both SOAP and REST APIs to partners using modern technology.

Responsibilities:

- Implemented common **Spring boot** components for [exception handling](#), error mapping, **GraphQL** client and custom **gradle plugins**, used across the microservices.
- Performance testing and optimization using Spring **Webflux**.

Sparkaufen	Technical Architect	May 2021 – Jan 2022	Neo Save
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Description: Sparkaufen digital piggy bank where you can earn money when you save to buy from your favourite brands

Responsibilities:

- Architecture, system and database design for managing Merchants, Stores, Products and Plans implemented using **Spring boot microservices**, Spring **Webflux** and **MongoDB**.
- Integrated with Paynetics digital banking APIs.

ClearCaptions	Technical Architect	Jan 2021 – Apr 2021	ClearCaptions
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Description: ClearCaptions is a Federal Communications Commission (FCC) certified telephone captioning provider. It enables anyone with hearing loss to communicate easier with the most advanced call captioning system available. It is a call captioning service that provides near real-time display of caller's words during phone conversations, a service intended for individuals whose hearing loss inhibits their phone use. The services are available either through ClearCaptions mobile app for both Android and iOS mobiles (VoIP calls) or landline phone devices (Bluephones).

Responsibilities:

- Architecture and database design from scratch following microservices patterns and DDD principles, **CQRS & ES** based implementation using **AXON** Server.
- Kafka Connect** for data migration.
- Implementation of E911 Service, User profiles and device management services.

PeaS	Technical Architect	Feb 2019 – Aug 2020	Unimoni
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Description: Pricing engine as a Service in Financial Service Industry is a Price discovery and management Product for inter and intra border transactions, facilitating money transfer across countries and currencies.

Responsibilities:

- Architecture and database design from scratch following **microservices** patterns and DDD principles and implementation using **Spring boot** microservices, **Spring Rest**, **Spring Reactor**, **Spring Data JPA**, **Hibernate**, **Server Sent Events (SSE)**, **ActiveMQ**, ensuring a scalable and robust system to handle complex financial transactions.
- Formulated and implemented integration strategies with rate providers such as FutureSource and Thomson Reuters. Ensured the architecture could handle high-volume transactions and maintained low latency in rate calculations
- Designed and implemented a flexible rate management system that supports various mark-up layers, including Interbank Rate, Value at Risk, Settlement, Agent, and Agent branch margin.
- Enabled dynamic pricing adjustments based on real-time data from integrated rate providers and rate sync with their legacy system.
- API based pricing services exposure to partners.
- Ensured all APIs and services adhered to the highest security standards and complied with relevant financial regulations.

- Led a team of developers, guiding them through the implementation of the architectural vision.
- Worked closely with stakeholders to understand business requirements providing technical solutions for the same.

Aperture	Team Lead	Jan 2018 – Jan 2019	Gspann
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Description: A Professional Services Automation product to manage Company Human resource's lifecycle, business accounts and projects, resource's allocations to projects as per SOW contracts, timesheets management, reports much etc.

Responsibilities:

- Implementation using **Spring boot microservices** following DDD principles, **Spring Data JPA**, **Hibernate**, **Spring security** and **Keycloak**. Unit and integration testing.
- Worked closely with stakeholders to understand business requirements and translate them into technical solutions.
- Led a team of developers, guiding them through the implementation of the architectural vision.

NDC Aggregator	Module Lead	Sep 2017 – Nov 2017	IGT
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Description: With the evolution of New Distribution Capability (NDC) standard by IATA, to distribute Air and Non-air products such as flight tickets and ancillary items, there is requirement for a mediator between Seller such as OTA/TMC and NDC compliant airlines. NDC aggregator serves just that mediator doing following activities.

Responsibilities:

- Requirement gathering, Architecture design and documentation, Development, Unit, Integration and performance testing.
- Implemented **Spring boot microservices** and **Integration patterns - Content based Router, Transformer, Multicast, Aggregator, Enricher** using **Apache Camel**.

VIS	Module Lead	Sep 2016 – Oct 2017	SITA
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Description: The Visitor Information System (VIS) is an integrated system that empowers various organizations to identify, monitor and manage visitors and travellers at border controls and within Sultanate of Oman, and includes an entry/exit system, Risk Manager, Watch lists, Targets, Actions, Profiler, Identity Manager, Event Manager

Responsibilities:

- **SOAP** Web services implementation using **Spring WS**, **EIPs** using **Apache Camel**.

SITA E-commerce	Module Lead	Sep 2014 – Aug 2016	SITA
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Description: The SITA E-Commerce product (SEC) provides a web-based application for the sale and servicing of bookings. The core areas of functionality provided are sale of Air and Car products, modification of Air bookings, Customer profiles, Delegated agent and a range of administration functions.

Educational Qualifications

DEGREE	SPECIALIZATION	COLLEGE/UNIVERSITY	PERCENTAGE	YEAR
B.E.	Computer Science and Engineering	ITM Gurgaon(MDU)	64%	2006
12 th	Non-Medical Science	HBSE(Haryana)	64%	2001
10 th	Math, Science, Social Science, English, Hindi	CBSE	71%	1999

Employment History

Period	Months	Organization	Role
April 2024 – Till date	17	Whilter.AI	Solution Architect
Feb 2022 – Nov 2023	20	StatusNeo Technology Consulting Pvt. Ltd.	Technical Architect
Feb 2019 – Jan 2022	35	Freelancing/Contracting	Technical Consultant
Feb 2018 – Jan 2019	11	Whisk Software Pvt. Ltd.(Gspann)	Technical Lead
Sep 2014 – Jan 2018	40	InterGlobe Technologies	Module Lead
May 2013 – Aug 2014	16	Mpurse Services Pvt. Ltd.	Assistant Manager
Jan 2012 – Apr 2013	16	Aricent Technologies (Holdings) Ltd.	Senior Software Engineer
Jun 2011 – Dec 2011	7	TCS	Business Associate
Dec 2009 – May 2010	6	IBM	Software Associate
Dec 2007 - Nov 2009	24	One97 Communications Pvt. Ltd	Senior Software Developer
Jun 2006 - Dec 2007	18	Meera Infotech Pvt. Ltd.	Software Engineer